

Lesson

7

Motivation

A problem well-put is half solved.

—JOHN DEWEY

Looking back, I think it was more difficult to see what the problems were than to solve them.

—CHARLES DARWIN

The formulation of a problem is often more essential than its solution, which may be merely a matter of mathematical or experimental skill. To raise new questions, new possibilities, to regard old questions from a new angle, requires creative imagination and marks real advance in science.

—ALBERT EINSTEIN

ACADEMIC AND PROFESSIONAL WRITING

The principles I discussed in Part Two apply to sentences and short passages in all kinds of writing. I now turn in Part Three to principles that apply to longer units of discourse, to sections and whole texts. While these principles are relevant to most kinds of expository and argumentative writing, I focus on how they apply to kinds of writing people do in college, as scholars and researchers, and on the job.

Of course, the various GENRES of academic and professional writing—critical essays, research reports, legal briefs, business proposals, even op-eds—differ from one another in many ways. But they also share features in common, features readers rely on to help them follow and make sense of what they read. This lesson addresses two of the most important of these: introductions and conclusions. Lesson 8 explains things you can do throughout your text to lead readers to experience it as organized and coherent. In both lessons, I take my examples mainly from academic genres, but the advice and principles apply to professional genres as well.

UNDERSTANDING MOTIVATION

Writers working in academic and professional contexts generally want to inform or persuade readers who know less than they do, which means they are vulnerable to thinking their writing is clear when their readers won't. Also, they can't assume their readers are already deeply invested in what they have to say. Accordingly, when you write in these contexts, you face a double challenge:

- to let readers know what to expect so that they can read more knowledgeably
- to motivate readers so that they want to read carefully

You do these things best when you write not just about a *topic* that interests you but about a *problem* that is important to your readers.

Stating Problems in Introductions

From the moment you begin to plan a writing project, think of your task not just as passing on information that interests *you*, but as posing a problem that will interest *your readers*. If they don't yet care—or even know—about that problem, you have to show them why they should. Such readers will ask, *So what?* That's a fair question, and you get one shot at answering it: in your introduction. That's where you must motivate readers to see your problem as their own.

For example, read this introduction (all these examples are much shorter than typical ones):

- 1a. When college students go out to relax on the weekend, many now "binge," downing several alcoholic drinks quickly until they are drunk or even pass out. It is a behavior that has been spreading through

colleges and universities across the country, especially at large state universities. It once was done mostly by men, but now even women binge. It has drawn the attention of parents, college administrators, and researchers.

That introduction offers only a topic: it does not motivate us to care about it. Unless a reader is already interested in the issue, she may shrug and ask, *So what? Who cares that college students drink a lot?*

Contrast that introduction with this one: it tells us why bingeing is not just an interesting topic but a problem worth our attention:

1b. Alcohol has been a big part of college life for hundreds of years. From football weekends to fraternity parties, college students drink and often drink hard. But a new kind of drinking known as "binge" drinking is spreading through our colleges and universities. Bingers drink quickly not to be sociable but to get drunk or even to pass out. Bingeing is far from the harmless fun long associated with college life. In the last six months, it has been cited in at least six deaths, many injuries, and considerable destruction of property. It crosses the line from fun to reckless behavior that kills and injures not just drinkers but those around them. We may not be able to stop bingeing entirely, but we must try to control its worst costs by educating students in how to manage its risks.

As short as that is, (1b) has the three parts that appear in most introductions. Each part has a role in motivating a reader to read on. The parts are these:

Shared Context—Problem—Solution/Main Point/Claim

Alcohol has been a big part of college life . . . drink hard. ^{shared context}
But a new kind of drinking known as "binge" drinking is spreading . . . kills and injures not just drinkers but those around them. ^{problem}
We may not be able to stop bingeing entirely, but we must try to control its worst costs by educating students in how to manage its risks. ^{solution/main point/claim}

Part 1: Establishing a Shared Context

Most pieces of writing open with a shared context, as does (1b):

Alcohol has been a big part of college life for hundreds of years. From football weekends to fraternity parties, college students drink and often drink hard. ^{shared context} But a new kind of drinking known as "binge" . . .

That shared context offers historical background, but it might have been a recent event, a common belief, or anything else that reminds readers of what they know, have experienced, or readily accept.

Event: A recent State U survey showed that 80% of first-year students engaged in underage drinking in their first month on campus, a fact that should surprise no one. ^{shared context} But what is worrisome is the spread among first-year students of a new kind of drinking known as "binge" . . .

Belief: Most students believe that college is a safe place to drink for those who live on or near campus. And for the most part they are right. ^{shared context} But for those students who get caught up in the new trend of "binge" drinking, . . .

These forms of shared context play a special role in motivating readers to read on. In (1b), I wanted you to accept that context as a seemingly unproblematic base for thinking about binge drinking *just so that I could then challenge it*. I set you up so that I could say, in effect, *you may think you know the whole story, but you don't*. That *but* signals the coming qualification:

. . . drink and often drink hard. ^{shared context} **BUT a new kind of drinking known as "binge" drinking is spreading . . .**

In other words, college drinking seems unproblematic, *but it turns out not to be*. I wanted that small surprise to motivate you to go on reading.

No opening move is more common among experienced writers: open with a seeming truth, then qualify or even reject it. You can find countless examples in newspapers, magazines, and especially professional journals. This opening context can be a sentence or two, as here; in a journal, it can be paragraphs long, where it is called a *literature review*, a survey of what researchers have said that the writer will qualify or correct.

Part 2: Stating the Problem

Sometimes when writers address specialized readers, they might skip the shared context and jump directly to the second part of an introduction: the statement of a problem. More commonly, though, they open with a shared context to set up this problem statement, introducing it with a word such as *but* or *however*:

Alcohol has been a big part of college life for hundreds of years. From football weekends to fraternity parties, college students drink and often drink hard. ^{shared context} But a new kind of drinking known as "binge" drinking is spreading through our colleges and universities. Bingers drink quickly not to be sociable but to get drunk or even to pass out. Bingeing is far from the harmless fun long associated with college life. In the last six months, it has been cited in at least

six deaths, many injuries, and considerable destruction of property. It crosses the line from fun to reckless behavior that kills and injures not just drinkers but those around them. ^{problem} We may not be able to . . .

The Two Parts of a Problem For readers to think that something is a problem, it must have two parts:

- The first part is some *condition* or *situation*: terrorism, rising tuition, binge drinking, anything that has the potential to cause trouble.
- The second part is the *intolerable consequence* of that condition, a *cost* that readers don't want to pay.

That cost is what motivates readers. They want to eliminate or at least ameliorate it, because it makes them unhappy. To identify the cost of a problem, imagine someone asking *So what?* after you state its condition. Answer that question and you have found the cost:

But a new kind of drinking known as "binge" drinking is spreading through our colleges and universities. Bingers drink quickly not to be sociable but to get drunk or even to pass out. ^{condition} *So what?* Bingeing is far from the harmless fun long associated with college life. In the last six months, it has been cited in at least six deaths, many injuries, and considerable destruction of property. It crosses the line from fun to reckless behavior that kills and injures not just drinkers but those around them. ^{cost of the condition}

The condition is binge drinking; the cost is death and injury. Readers have to see the condition and cost *together* before they recognize a problem.

Two Kinds of Problems: Practical and Conceptual There are two kinds of problems, and each motivates readers in a different way. You have to write about them differently.

- A *practical* problem concerns a condition or situation in the world and demands an *action* as its solution. That students binge drink and harm themselves is a practical problem.
- A *conceptual* problem concerns what we think about something and demands a *change in understanding* as its solution. That we don't know why students binge is a conceptual problem.

Writers outside the academic world most often address practical problems; writers inside it most often address conceptual ones.

Practical Problems We usually name a practical problem in a word or two: *cancer, unemployment, binge drinking*. But that's a shorthand. Those terms name only the condition: they say nothing about costs. Most conditions sound like trouble, but anything can be the condition of a problem if its palpable costs make you unhappy. If winning the lottery made you suffer the loss of friends and family, it would be a practical problem.

You may think that the costs of a problem like bingeing are too obvious to state, but you cannot count on readers to see the problem as you do. Some readers may see different costs. Where you see death and injury, a university publicist might see only bad press: *Those binge drinking students make us look like a party school, which hurts our image with parents*. More callous readers might see no costs at all: *So what if college students injure or kill themselves? What's that to me?* If so, you have to figure out how to make such readers see that those costs affect them. If you can't describe the costs you see so that they matter *to your readers*, they won't care about what you've written.

Conceptual Problems A conceptual problem has the same two parts as a practical one: a condition and its costs. But beyond that, the two sorts of problem are very different.

The condition of a conceptual problem is always something that we do not know or understand. We can express this condition as a question: *How much does the universe weigh? Why does the hair on your head keep growing, but the hair on your legs doesn't?*

The cost of a conceptual problem is not the palpable unhappiness we feel from pain, suffering, and loss; it is the dissatisfaction we feel because we don't understand something important to us. We can express this cost as something *more important* that readers don't know, as *another, larger question*:

Cosmologists do not know how much the universe weighs. ^{condition} *So what?* Well, if they knew, they might figure out something more important: Will time and space go on forever, or will they end? And if they do, when and how? ^{cost/larger question}

Biologists don't know why some hair keeps growing and other hair stops. ^{condition} *So what?* If they knew, they might understand something more important: What turns growth on and off? ^{cost/larger question}

Administrators do not know why students underestimate the risks of binge drinking. ^{condition} *So what?* If they knew, they might figure out something more important: Would better information at orientation help students make safer decisions about drinking? ^{cost/larger question}

Sometimes, as in the last example, the larger question is about something readers do not know how to do. But that is still a conceptual problem, because it concerns our ignorance and because its solution is not an action but information.

Think of it like this: for a conceptual problem, you answer a small question so that your answer contributes to answering a larger, more important one. Readers are motivated because your small question inherits its importance from that larger one.

Here's the point: Like your readers, you will usually be more motivated by large questions. But limited resources—time, funding, knowledge, skill, pages—may keep you from addressing a large question satisfactorily. So you have to find a question you *can* answer. When you plan your paper, look for a question that is small enough to answer but is also connected to another question large enough for you *and your readers* to care about.

Part 3: Stating the Solution

The solution is your main point or claim. We solve practical problems with action: readers (or someone) must *change what they do*. We solve conceptual problems with information: readers (or someone) must *change what they think*. Your answer to a small question then helps readers understand a larger one.

Practical Problems: What We Should Do To solve a practical problem, you must propose that the reader (or someone) *do* something to change a condition in the world:

... behavior that crosses the line from fun to recklessness that kills and injures not just drinkers but those around them. ^{problem} **We may not be able to stop bingeing entirely, but we must try to control its worst costs by educating students in how to manage its risks.** _{solution/point}

Conceptual Problems: What We Should Think To solve a conceptual problem, you must state something you want readers to *understand* or *believe*:

... we can better understand not only the causes of this dangerous behavior but also the nature of risk-taking behavior in general. ^{problem} **This study reports on our analysis of the beliefs of 300 first-year**

college students. We found that students were more likely to binge if they knew more stories of other students bingeing, so that they believed that bingeing is far more common than it actually is. _{solution/point}

As Darwin and Einstein said, nothing is more difficult than finding a good question, because without one, you don't have an answer worth supporting.

Here's the point: Some writers, especially student writers, believe the impact of their ideas depends on how forcefully they state them. But a strong main point, claim, or thesis alone will not make a paper persuasive. Readers will find your claims most meaningful if you present them as solutions to problems, practical or conceptual, that your readers care about.

QUICK TIP When you read an academic book or article, look first for the implied question in its problem statement and then for its main claim, which answers that question. They will help focus your reading. If you don't find a question in the introduction, look for one in the conclusion. If that fails, find the main claim and ask yourself, *What question does this answer?* The more you understand *why* a writer is telling you something, the better you will understand what she writes.

Another Part: Prelude

What best motivates readers is a problem in need of a solution, but a catchy opening can vividly introduce themes central to your problem. To name this device, we can use a musical term: *prelude*. Writers in the natural and social sciences rarely use preludes. They are more common in the humanities and most common in writing for the general public.

Here are three preludes that could establish key themes in a paper about binge drinking.

1. A Quotation

"If you're old enough to fight for your country, you're old enough to drink to it."

2. A Startling Fact

A recent study reports that at most colleges three out of four students “binged” at least once in the previous thirty days, consuming more than five drinks at a sitting. Almost half binge once a week, and those who binge most are not just members of fraternities but their officers.

3. An Illustrative Anecdote

When Jim S., president of Omega Alpha, accepted a dare from his fraternity brothers to down a pint of whiskey in one long swallow, he didn’t plan to become this year’s eighth college fatality from alcohol poisoning.

We can combine all three:

It is often said that “if you’re old enough to fight for your country, you’re old enough to drink to it.”^{quotation} Tragically, Jim S., president of Omega Alpha, no longer has a chance to do either. When he accepted a dare from his fraternity brothers to down a pint of whiskey in one long swallow, he didn’t expect to become this year’s eighth college fatality from alcohol poisoning.^{anecdote} According to a recent study, at most colleges, three out of four students have, like Jim, drunk five drinks at a sitting in the last thirty days. And those who drink the most are not just members of fraternities but—like Jim S.—officers.^{striking fact}

Drinking, of course, has been a part of American college life since the first college opened . . .^{shared context} But in recent years . . .^{problem}

Here, then, is a general plan for your introductions:

Prelude
Shared Context
Problem [Condition + Cost]
Solution / Main Point / Claim

Not every introduction must follow that plan exactly. Preludes are relatively uncommon. In special circumstances, writers omit or rearrange other parts as well. Here are four common variations:

- When readers know a subject well, writers sometimes skip the shared context and open with the problem.
- When readers know a problem well, writers sometimes state the cost before the condition.

- When readers can be counted on to recognize the costs, writers sometimes skip them and just state the condition.
- When readers are willing to wait (which is rare), writers sometimes save their solution/main point/claim for the conclusion, ending the introduction with a promise that a solution will come.

Although many successful introductions use these variations, your best option is to follow the general pattern until you have lots of experience with it.

HOW TO REVISE: INTRODUCTIONS

To analyze, assess, and revise your introduction, do this:

1. **Determine whether you are posing a practical or conceptual problem.** Do you want readers to *do* something or to *think* something?
2. **Draw a line after your introduction.** If you cannot quickly locate the end of your introduction, neither will your readers, who might then miss both your problem and its solution, the main point of your paper.
3. **Divide the introduction into its three parts: shared context + problem + solution/main point/claim.** If you cannot quickly make those divisions, your introduction is likely to seem unfocused.
4. **Make sure the first word of the first sentence after the shared context is *but*, *however*, or some other word indicating that you will challenge that shared context.** If you don’t explicitly signal the contrast between the shared context and the problem, readers may miss it.
5. **Divide the problem into two parts: condition and cost.**
 - 5a. **Is the condition the right kind for the problem?**
 - For a practical problem, the condition must be something that exacts a palpable cost.
 - For a conceptual problem, the condition must be something not known or understood. This should be stated not as a direct question, *What causes bingeing?*, but as a statement of what we do not know: *But we do not know why bingers ignore known risks.*

5b. Does the cost appropriately answer *So what?*

- For a practical problem, the answer to *So what?* must state some palpable consequence of the condition that causes unhappiness.
- For a conceptual problem, the answer to *So what?* must state some more significant issue that is not known or understood.

6. **Underline your solution/main point/claim.** This should appear at the end of the introduction in its stress position, and should state the key themes that the rest of your paper will develop (more on that in Lesson 8).

CONCLUSIONS

A good introduction motivates your readers, introduces your key themes, and states your main point, the solution to your motivating problem. Get your introduction straight, and readers can read the rest more quickly and understand it better.

A good conclusion serves a different end: as the last thing your reader reads, it should bring together your point, its significance, and its implications for thinking further about your problem. Conclusions vary more than introductions, but in a pinch, you can map the parts of your introduction onto your conclusion. Just reverse their order:

1. **Open your conclusion by stating (or restating) the gist of your point, the main claim of your paper, the solution to your problem:**

Though we can come at the problem of bingeing from several directions, the most important is education, especially in the first week of a student's college life. But that means each university must devote time and resources to it.

2. **Explain its significance by answering *So what?* Answer in a new way, if you can; if you can't, restate what you offered in the introduction, now as a benefit:**

If we do not start to control bingeing soon, many more students will die.

If we start to control bingeing now, we will save many lives.

3. **Suggest a further question or problem to be resolved, something still not known. Answer *Now what?*:**

Of course, even if we can control bingeing, the larger issue of risk-taking in general will remain a serious problem.

4. **End with an anecdote, quotation, or fact that echoes your prelude. We'll call this by another musical term, your *coda* (again, used most often in popular writing, rarely in the natural and social sciences):**

We should not underestimate how deeply entrenched bingeing is: We might have hoped that after Jim S.'s death from alcohol poisoning, his university would have taken steps to prevent more such tragedies. Sad to say, it reported another death from bingeing this month.

There are other ways to conclude, but this one works when nothing better comes to mind.

IN YOUR OWN WORDS**Exercise 7.1**

For this exercise, you can use pieces of writing that are finished or still in progress. For each, draw a line between the problem and the solution. Underline the condition and bracket the cost. Classify the problem of each as conceptual or practical. Then rewrite the conceptual problems as practical ones and the practical problems as conceptual ones. What did you have to change? Were some problems easier to rewrite in this way than others? Why?

Exercise 7.2

Writers, especially in academic contexts, can struggle more with problem statements than with other parts of their introductions. It makes sense that writers would have this struggle. To state a problem, writers need to understand not only their own ideas but also the motives of their readers. This exercise will help you do that. For a project you are just beginning, do the following:

1. In a sentence, state your *topic*: In this paper I am writing about _____ (e.g., science in Tennyson's *In Memoriam*, the effects of *Citizens United v. Federal Election Commission* on election financing, the space shuttle *Columbia* accident).
2. In a sentence, state why *you* selected this particular topic out of the many you could have chosen: I care about my topic because _____. (If you cannot complete this sentence, you need to do more thinking about your topic or choose another. You cannot expect your readers to care if you can't say why you do.)
3. In a sentence, state your *main point* or *claim*: The point I want to make about [topic] is that _____.

4. In a paragraph, describe one person (real or imagined) who would care deeply about your ideas. What does she (or he) look like? What does she do for a living? What are her personal interests? What books has she read in the past two months?
5. Now write a short letter to the person you have described, explaining to her why she should care about your paper.

You will probably find phrases and ideas in this letter that you can use in your problem statement.

Exercise 7.3

Analyze introductions from three readings in your area of study or expertise using the procedure on pp. 103–104:

1. Determine whether the problem is practical or conceptual.
2. Draw a line at the introduction's end.
3. Divide the introduction into its three parts: shared context, problem, and solution/main point/claim. Do these three parts appear? Are they in the right order?
4. Circle the word or phrase (usually *but* or *however*) that indicates a challenge to the shared context.
5. Divide the problem into its two parts: condition and cost. Does the cost answer the question *So what?*
6. Underline the solution/main point/claim.

What patterns did you notice? Now do the same with an introduction of your own.

SUMMING UP

You best motivate readers with an introduction that states a problem readers want to see solved.

For a practical problem the key is to state its costs so clearly that readers will ask not *So what?* but *What do we do?* Here is a plan for introducing a practical problem:

Alcohol has been a part of college life for hundreds of years. From football weekends to fraternity parties, college students drink and often drink hard. shared context

Open the introduction with *shared context*, a brief statement of what you will go on to qualify or even contradict.

But a new kind of drinking known as "binge" drinking is spreading through our colleges and universities. Bingers drink quickly not to be sociable but to get drunk or even to pass out. condition *So what?*

Follow that with a statement of the condition of the problem. Introduce it with a *but*, *however*, *on the other hand*, etc. Imagine a *So what?* after it.

Bingeing is far from harmless. In the last six months, it has been cited in six deaths, many injuries, and considerable destruction of property. It crosses the line from fun to reckless behavior that kills and injures not just drinkers but those around them. costs

Answer that imagined *So what?* with a statement of the consequences of that condition, its costs *to your readers* that they do not want to pay.

We may not be able to stop bingeing entirely, but we must try to control its worst costs by educating students in how to manage its risks. solution

Conclude with a statement of the solution to the problem, an *action* that will eliminate or at least ameliorate the costs.

For conceptual problems, the key is to state a small question worth answering because it helps to answer a larger, more significant one. Here is a plan for introducing conceptual problems:

Colleges are reporting that binge drinking is increasing. We know its practical risks. We also know that bingers ignore those risks, even after they have learned about them. shared context

Open the introduction with *shared context*, a brief statement of what you will go on to qualify or even contradict.

But we don't know what causes bingers to ignore the known risks: social influences, a personality attracted to risk, or a failure to understand the nature of the risks.

condition/first, small question *So what?*

If we can determine why bingers ignore known risks of their actions, we can better understand not only the causes of this dangerous behavior but also the nature of risk-taking behavior in general.

cost/second, larger question

In this study, we analyzed the beliefs of 300 first-year college students to determine . . . We found that . . .

solution

Follow that with a statement of the condition of the problem. Introduce it with a *but*, *however*, *on the other hand*, etc. State something that is not known or well understood. Imagine a *So what?* after it.

Answer that imagined *So what?* with the cost of the condition, a larger and more important issue that is not known or understood but that might be answered if we know the answer to the first question.

Conclude your introduction with a statement of the solution to the problem, an answer to the first question that helps answer the second one as well.

Lesson

8

Global Coherence

One of the most difficult things [to write] is the first paragraph. I have spent many months on a first paragraph, and once I get it, the rest just comes out very easily. In the first paragraph you solve most of the problems with your book. The theme is defined, the style, the tone.

—GABRIEL GARCÍA MÁRQUEZ

UNDERSTANDING GLOBAL COHERENCE

In the last lesson, I explained how to create an introduction that does two things:

- Motivates your readers by stating a problem that they care about.
- Frames the rest of your document by stating the point and key concepts that you will develop in what follows.

In this lesson, I explain how that second point applies to all the parts of your document—its sections, subsections, and even paragraphs. Like the term *clear*, the term *coherent* doesn't refer to anything we find on the page. Coherence is an experience we create for ourselves as we make sense out of what we read.

What we look for on the page are signals that help us integrate what we are reading with the knowledge we already have. Your readers will understand your writing better and more easily when you build those signals into it deliberately. This lesson explains how to do that.

FORECASTING THEMES TO CREATE COHERENCE

In Lessons 5 and 6, we looked at those features that help readers create *local* coherence in short passages, but readers need more to grasp the *global* coherence of a whole document. To help them, you can use a by-now-familiar principle: begin each section or subsection of a document with a short, easily grasped segment that states its point and introduces the themes that organize the longer segment that follows, the body. Then in that body, support, develop, or explain that point and those themes.

To help readers grasp the coherence of a document and its sections, follow these six principles:

For the document:

1. Readers must know where the introduction ends and the body begins, as well as where each section ends and the next begins. Identify the start of each new section with a heading that includes the key themes for that section (see 5 below). If your field does not use headings, delete them for the final draft.
2. At the end of the introduction, readers look for the document's main point, claim, or solution to its problem, which should state the main themes developed in the rest. If you have good reason to save your main point for the conclusion, end your introduction with a sentence that promises the point to come *and* states the main themes.
3. In the body, readers look for the concepts announced as themes at the end of the introduction, using them to organize their understanding of the whole. Be sure that you repeat those themes regularly.

For each section and subsection:

4. Readers look for a short segment that introduces the section or subsection.
5. At the end of that introductory segment, readers look for a sentence that states both the point of the section and the specific concepts you will develop as distinctive themes for that section.
6. In the body of the section, readers look for the concepts announced as themes at the end of the introductory segment, using them to organize their understanding of that section. Be sure that you repeat them regularly.

QUICK TIP You can use these six principles to prepare yourself to read a difficult document. First, highlight the question in the problem statement and the main claim that answers it (see pp. 96–101). Next, for each section, highlight its introduction, point, and key concepts. If you don't find them in the introduction to a section, look for them at the end of the section. Finally, read through just the parts that you highlighted. When you then begin reading in detail, you will have in mind an overview that will help you better understand and remember the rest.

In the limited space we have here, I can't illustrate these principles with entire documents or even long sections. So I will use paragraphs and ask you to relate their structure to that of a whole section of a document.

For example, read this:

1a. Thirty sixth-grade students wrote essays that were analyzed to determine the effectiveness of eight weeks of training to distinguish fact from opinion. That ability is an important aspect of making sound arguments of any kind. In an essay written before instruction began, the writers failed almost completely to distinguish fact from opinion. In an essay written after four weeks of instruction, the students visibly attempted to distinguish fact from opinion, but did so inconsistently. In three more essays, they distinguished fact from opinion more consistently, but never achieved the predicted level of performance. In a final essay written six months after instruction ended, they did no better than they did in their pre-instruction essays. Their training had some effect on their writing during the instruction period, but it was inconsistent, and six months after instruction it had no measurable effect.

The first few sentences introduce the rest, but we don't see in them the key concepts that follow: *inconsistently*, *never achieved*, *no better*, *no measurable effect*. Those terms are crucial to the *point* of the whole passage. Worse, the passage doesn't give us that point until the very end: training had no long-term effect. And so as we read, the passage seems to ramble, until the end, when we learn what we need to know to make sense of it retrospectively. But that takes more effort than we should have to expend.

Compare this version:

1b. In this study, thirty sixth-grade students were taught to distinguish fact from opinion. They did so successfully during the instruction period, but the effect was inconsistent and less than predicted, and six

months after instruction ended, the instruction had no measurable effect. In an essay written before instruction began, the writers failed almost completely to distinguish fact from opinion. In an essay written after four weeks of instruction, the students visibly attempted to distinguish fact from opinion, but did so inconsistently. In three more essays, they distinguished fact from opinion more consistently, but never achieved the predicted level of performance. In a final essay written six months after instruction ended, they did no better than they did in their pre-instruction essay. We thus conclude that short-term training to distinguish fact from opinion has no consistent or long-term effect.

In (1b), we quickly grasp that the first two sentences introduce what follows. And in the second sentence, we see two things: both the point of the passage (underlined) and its key terms (**boldfaced**):

1b. In this study, thirty sixth-grade students were taught to distinguish fact from opinion. They did so **successfully** during the instruction period, but the effect was **inconsistent** and **less than predicted**, and six months after instruction ended, the instruction had no measurable effect.

Consequently, we feel the passage hangs together better, and we read it with more understanding.

Now imagine two documents: in one, the point of each section and of the whole appears at its *end* (as in 1a) and what openings there are do not introduce the key terms that follow; in the other, each point appears in an *introductory* segment to every paragraph, section, and of the whole (as in 1b). Which would be easier to read and understand? The second, of course.

Keep in mind this principle: put the point sentence at the end of the short opening segment; make it the *last* sentence that your reader reads before starting the longer, more complex segment that follows.

- In a paragraph, the introductory segment might be just a single sentence, so by default, it will be the last sentence readers read before they read what follows. If the paragraph has a *two-sentence* introduction (as did 1b), be sure that its point is the *second* sentence, still making it the last thing readers read before they read the rest.
- For sections, your introduction might be a paragraph or more. For a whole document, you might need several paragraphs. Even in those cases, put your point sentence at the end of that introductory segment, no matter how long it is. Make your point the last thing readers read before they begin reading the longer, more complex segments that follow.

Some inexperienced writers think that if they reveal their main point in their introduction, readers will be bored and not read on. Not true. If you motivate readers with an interesting problem, they will want to see how you address it.

Here's the point: To write a document that readers will think is coherent, open every unit—section, subsection, and the whole—with a short, easily grasped introductory segment. At the end of that opening segment, put a sentence that states both the point of the unit and the key concepts that follow. Such “point” sentences constitute the outline of your document, its logical structure. If readers miss them, they may judge your writing to be incoherent.

TWO MORE REQUIREMENTS FOR COHERENCE

We can make sense of almost anything we read if we know its points. But to make full sense of a passage, we must see two more things.

1. **Readers must see how everything in a section or whole is relevant to its point.** Consider this passage:

We analyzed essays written by sixth-grade students to determine the effectiveness of training in distinguishing fact from opinion. In an essay written before training, the students failed almost completely to distinguish fact and opinion. These essays were also badly organized in several ways. In the first two essays after training began, the students attempted to distinguish fact from opinion, but did so inconsistently. They also produced fewer spelling and punctuation errors. In the essays four through seven, they distinguished fact from opinion more consistently, but in their final essay, written six months after completion of instruction, they did no better than they did in their first essay. Their last essay was significantly longer than their first one, however. Their training thus had some effect on their writing during the training period, but it was inconsistent and transient.

What are those sentences about spelling, organization, and length doing there? When readers can't see the relevance of sentences to a point, they are likely to judge what they read as being incoherent.

Unfortunately, I can't give you a simple rule of relevance, because it's so abstract a quality. I can only list its most important kinds. Sentences are relevant to a point when they offer these:

- background or context
- points of sections and the whole
- reasons supporting a point
- evidence, facts, or data supporting a reason
- an explanation of reasoning or methods
- consideration of other points of view

2. **Readers must see how the parts of your document are ordered.** Readers want to see not just how everything they read is relevant to a point, but what principle is behind the order of its parts. We look for three kinds of order: chronological, coordinate, and logical.

- **Chronological** This is the simplest order, from earlier to later (or vice versa), as a narrative or as cause and effect. Signal time with *first, then, finally*; signal cause and effect with *as a result, because of that*, and so on. The passage about the essay research was chronologically organized.
- **Coordinate** Two or more sections are coordinate when they are like pillars equally supporting a common roof. *There are three reasons why. . .* Order those sections so that their sequence makes sense to your reader—by importance, complexity, and so on—then signal that order with words and phrases such as *first, second, . . .* or *also, another, more important, in addition*, and so on. That's how this section on order is organized.
- **Logical** This is the most complex order: by example and generalization (or vice versa), premise and conclusion (or vice versa), or by assertion and contradiction. Signal logic with *for example, on the other hand, it follows that. . .*

QUICK TIP Writers often order their documents chronologically because that is easiest for them. Once you have drafted a paper, read it through to see whether you have organized it simply as a narrative of your thinking. If you have, consider revising. Order your ideas not in the way that is easiest for you, but in the way that best helps your readers understand them.

ON PARAGRAPHS

In different kinds of writing, paragraphs follow different conventions. In newspaper articles, they are often only a sentence long. In this book, they are a bit longer. In academic papers and scholarly journal articles, they can run half a page or more. Here is some advice that will help you write coherent longer paragraphs:

- Begin with one or two short, easily grasped sentences that frame what follows.
- State the point of the paragraph (in traditional terms its *topic sentence*) in the last sentence of its introduction. If the introduction is just one sentence, it will be its point by default.
- Toward the end of that point sentence, name the key themes that thread through what follows.

Treat this advice as general guidance, not as a rigid template. Many longer paragraphs don't follow this tidy structure, and we get through them just fine. Even so, most begin with some kind of opening segment that frames what follows by introducing key themes and perhaps by stating the point. If the point doesn't appear at the beginning of a paragraph, it will usually come at the end.

Compare these two examples:

2a. The team obtained exact sequences of fossils—new lines of antelopes, giraffes, and elephants developing out of old and appearing in younger strata, then dying out as they were replaced by others in still later strata. The most specific sequences they reconstructed were several lines of pigs that had been common at the site and had developed rapidly. The team produced family trees that dated types of pigs so accurately that when they found pigs next to fossils of questionable age, they could use the pigs to date the fossils. By mapping every fossil precisely, the team was able to recreate exactly how and when the animals in a whole ecosystem evolved.

2b. By precisely mapping every fossil they found, the team was able to recreate exactly how and when the animals in a whole ecosystem evolved. They charted new lines of antelopes, giraffes, and elephants developing out of old and appearing in younger strata, then dying out as they were replaced by others in still later strata. The most exact sequences they reconstructed were several lines of pigs that had been common at the site and had developed rapidly. The team produced family trees that dated types of pigs so accurately that when they found pigs next to fossils of questionable age, they could use the pigs to date the fossils.

Paragraph (2a) makes its point in the last sentence; paragraph (2b) in its first sentence. Reading these paragraphs in isolation, you probably found (2b) slightly easier to understand. But in the context of an otherwise coherent text about fossil hunters and their work, (2a) probably wouldn't give you a problem.

If you have framed and organized your document and its sections well, your readers can make their way through a few paragraphs that are less than perfect. But if they don't know what your paragraphs are supposed to add up to, then no matter how well written they are individually, your readers may well feel lost.

A GENERAL PRINCIPLE OF CLARITY

This general principle implies many of our others. It applies to individual sentences, to longer paragraphs, to sections and subsections, and to whole documents:

Readers are more likely to judge as clear any unit of writing that opens with a short segment that they can easily grasp and that frames the longer and more complex segment that follows.

Sentences: In a simple sentence, that short, easily grasped segment is a subject/topic. Compare these two:

1a. Resistance in Nevada against its use as a waste disposal site has been heated.

1b. Nevada has heatedly resisted its use as a waste disposal site.

In a complex sentence, that short, easily grasped segment is a main clause that expresses the point of its sentence. Compare these two:

2a. Greater knowledge of pre-Columbian civilizations and the effect of European colonization destroying their societies by inflicting on them devastating diseases has led to a historical reassessment of Columbus's role in world history.

2b. Historians are reassessing Columbus's role in world history because they know more about pre-Columbian civilizations and how European colonization destroyed their societies by inflicting on them devastating diseases.

The point of sentence (2a) is buried at its end. In (2b), the opening clause states the main point of the sentence, its most important claim: *Historians are reassessing Columbus's role in world history*. That claim is then supported by the longer and more complex clause that follows.

Paragraphs: In a paragraph, that short, easily grasped unit is an introductory sentence or two that both expresses the point of the paragraph and introduces its key concepts. Compare these two paragraphs:

3a. Thirty sixth-grade students wrote essays that were analyzed to determine the effectiveness of eight weeks of training to distinguish fact from opinion. That ability is an important aspect of making sound arguments of any kind. In an essay written before instruction began, the writers failed almost completely to distinguish fact from opinion. In an essay written after four weeks of instruction, the students visibly attempted to distinguish fact from opinion, but did so inconsistently. In three more essays, they distinguished fact from opinion more consistently, but never achieved the predicted level. In a final essay written six months after instruction ended, they did no better than they did in their pre-instruction essay. Their training had some effect on their writing during the instruction period, but it was inconsistent, and six months after instruction it had no measurable effect.

3b. In this study, thirty sixth-grade students were taught to distinguish fact from opinion. They did so successfully during the instruction period, but the effect was inconsistent and less than predicted, and six months after instruction ended, the instruction had no measurable effect.

opening segment/point In an essay written before instruction began, the writers failed almost completely to distinguish fact from opinion. In an essay written after four weeks of instruction, the students visibly attempted to distinguish fact from opinion, but did so inconsistently. In three more essays, they distinguished fact from opinion more consistently, but never achieved the predicted level. In a final essay written six months after instruction ended, they did no better than they did in their pre-instruction essay. We thus conclude that short-term training to distinguish fact from opinion has no consistent or long-term effect.

Paragraph (3a) has no clearly distinguished opening unit, and it does not announce the key themes of the paragraph. Paragraph (3b) has a clearly marked opening unit that states the point, and it clearly announces the key themes of the paragraph.

Sections: In a section or subsection, that short, easily grasped unit may be just a paragraph; in longer units, it will be proportionally longer. Even so, at its end it expresses the point of its unit and introduces the key concepts that follow. There is not enough space here to illustrate how that principle applies to a passage several paragraphs long, but it is easy to imagine.

Whole documents: In a whole document, that introductory unit might be one or more paragraphs long, perhaps even a few pages. Even so, it should be substantially shorter than the rest, and in a sentence at its end, it should state the point of the whole document and introduces its key concepts.

QUICK TIP Budget your time for both drafting and revision so that you spend most of it on beginnings: the introduction to the whole, then the introductions to major sections, then introductions to subsections and long paragraphs, then the beginnings of sentences. Get beginnings straight, and the rest is likely to take care of itself.

The Costs and Benefits of Templated Writing

Some writers fear that patterns like these will inhibit their creativity and bore their readers. That's a reasonable concern, if you are writing a literary essay that explores your own thoughts as you have them, for readers who have the time and patience to follow the twists and turns of your thinking. If you are writing that kind of essay for that kind of reader, go to it. Don't tie yourself to what I've said here.

On most occasions, however, most of us read less for aesthetic pleasures than to understand what we need to know. You help readers toward this end when you follow the principles of clarity and coherence we've looked at in Parts Two and Three of this book.

Such writing may seem formulaic—to *you*, because *you* will be so conscious of the patterns you followed. But it earns the gratitude of readers who have too little time to read, understand, and remember everything they must and who will, in any event, focus more on understanding the substance of your writing than on critiquing its form.

IN YOUR OWN WORDS

Exercise 8.1

A basic principle of clarity is that any unit of discourse—a sentence, a paragraph, a section, a whole document—should begin with a short segment that introduces and frames the longer and more

complex segment that follows. Go through a piece of your writing section by section. Draw a line after that short segment and circle words in that segment that signal key themes in what follows. If you cannot, revise.

Exercise 8.2

To feel a document (or a section of one) is coherent, a reader needs to understand how it is organized (review pp. 113–114). But writers, especially in early drafts, often organize their documents in the way that is easiest for them, not the way that is best for their readers. Specifically, writers often adopt a chronological or narrative structure by default. You can see why this is so. When getting ideas down on paper, it is easiest for writers simply to rehearse their thinking or research. But most often, readers do not want to hear a story of discovery but to understand a writer's points. Revising for global coherence, therefore, often involves translating a document from a chronological or narrative structure to a coordinate or logical one. A reader can help you do this.

Go through a document or section that you have organized chronologically. Highlight your points, paragraph by paragraph, and copy them onto index cards. Shuffle the cards and give them to a reader. Have a reader put the cards into what seems like their right order. Reorganize your document or section so that it follows that order. What did you have to change?

SUMMING UP

Plan your paragraphs, sections, and the whole on this model:

Researchers have made strides in the **early and accurate diagnosis** of *Alzheimer's*, [but **those diagnoses** have raised A NEW HUMAN PROBLEM about **informing those at risk** before they show any **symptoms** of it.]_{point}

Open each unit with a relatively short segment introducing it.

End that segment with a sentence stating the point of that unit.

Toward the end of that point sentence, use key themes that the rest of the unit develops.

Not too long ago, when **physicians** examined an older patient who seemed *out of touch with reality*, they had to **guess** whether that person had *Alzheimer's* or was *only senile*. In the past few years, however, they have been able to use **new and more reliable tests** focusing on genetic clues. But in **the accuracy of these new tests** lies the RISK OF ANOTHER KIND OF HUMAN TRAGEDY: physicians may be able to **predict** *Alzheimer's* long before its overt appearance, but such an **early diagnosis** could PSYCHOLOGICALLY DEVASTATE AN APPARENTLY HEALTHY PERSON.

In the longer segment that follows, use consistent topics (underlined).

Repeat key terms introduced toward the end of the opening segment (**boldfaced**, *italicized*, and capitalized).

Make every sentence follow the old-new principle.

Order sentences, paragraphs, and sections in a way that readers understand.

Make all sentences relevant to the point of the unit that they constitute.

PART FOUR

Grace

There are two sorts of eloquence; the one indeed scarce deserves the name of it, which consists chiefly in laboured and polished periods, an over-curious and artificial arrangement of figures, tinselled over with a gaudy embellishment of words.... The other sort of eloquence is quite the reverse to this, and which may be said to be the true characteristic of the holy scriptures; where the eloquence does not arise from a laboured and farfetched elocution, but from a surprising mixture of simplicity and majesty.

—LAURENCE STERNE